

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- No Cummins® service tools required.

Additional Service Items

- No additional service items required.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

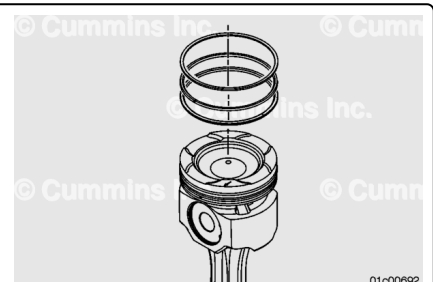
- Disconnect the batteries. See equipment manufacturer service information.
- Remove the piston and connecting rod assemblies from the engine. Refer to Procedure 001-054 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-054.html>)

Remove

⚠ CAUTION ⚠

Do not over-expand the second piston ring on pistons.

Use a universal ring expander to remove the second piston ring on the pistons.



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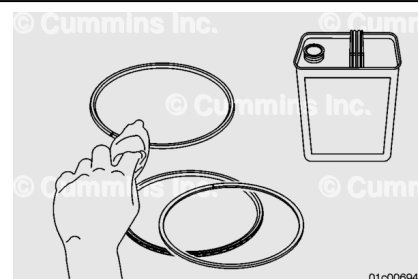
Place a tag on the rings, and record the cylinder number of the piston on the tag for future reference, if required.



Clean and Inspect for Reuse

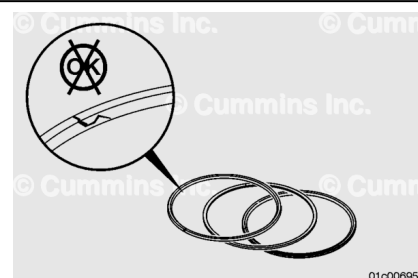
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Clean the rings with mineral spirits.

Inspect all of the piston rings for any cracks and chips on the ring face that mates with the liner. Discard and replace any rings with this type of damage.



Measure

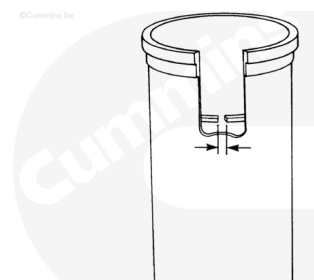
To check the ring gap, install the piston rings below the ring reversal area in the cylinder liner in which they were used.

Use the top part of the piston to correctly position the ring in the liner.



Use a feeler gauge to inspect the ring end gap.

Piston Ring Gap			
	mm		in
Top	0.23	MIN	0.009
	0.35	MAX	0.014
Second	1.87	MIN	0.074
	2.20	MAX	0.087
Oil	0.13	MIN	0.005
	0.35	MAX	0.014



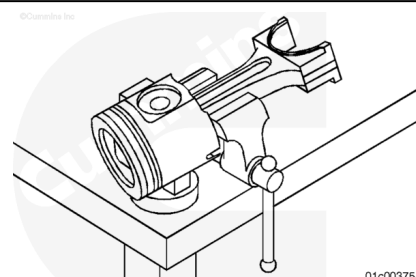
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If the piston ring gaps do **not** meet these specifications, the three piston rings **must** be replaced. Cummins Inc. recommends inspecting the piston and liner, and to consider replacement, if the piston and liner inspection does **not** meet specifications.

Install

⚠ CAUTION ⚠

Use a copper or aluminum vise protector when installing the connecting rod in a vise to protect the connecting rod from damage.



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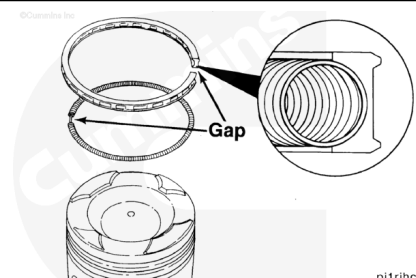
When reusing piston rings, be sure that the rings are matched up with the same piston from which they were removed.

The ring gap of each ring **must not** be aligned with the piston pin or with any other ring. If the ring gaps are **not** aligned correctly, the rings will **not** seal properly.

Install the connecting rod in a vise to hold the assembly in a horizontal position while installing the rings.

A cross-sectional view of an oil control ring is shown.

The two-piece oil control ring **must** be installed with the expander ring gap 180 degrees from the gap of the oil ring. Do **not** overlap the ends of the expander ring.



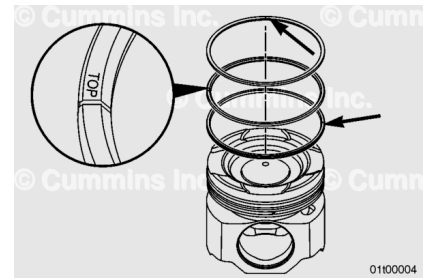
pi1rihc

All piston rings (top, second, and oil) have the top marked for correct installation.

Install the piston rings with TOP toward the top of the piston.

Note : All piston rings have the top marked for correct installation orientation.

Use a universal piston ring expander to install the second and the top rings into the piston.

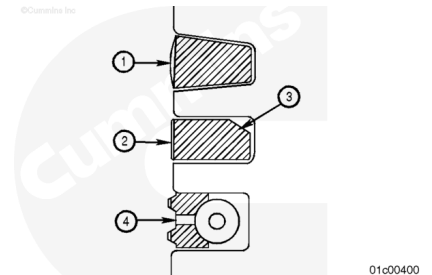


The piston **must** be held in a horizontal position to install the rings in order to keep the rings in the groove while installing the ring compressor.

The top piston ring (1) is a keystone ring with no twist or cutback notch.

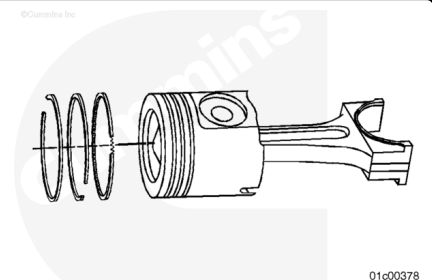
The second ring has a rectangular cross-section (2) with an angled cut (3) on the upper side of the inner diameter.

The oil control ring (4) is the bottom piston ring.



Rotate the rings to position the ring gaps as shown.

The ring gap of each ring **must not** be aligned with the piston pin, or with any other ring gap. If the ring gaps are **not** aligned correctly, the rings will **not** seal properly.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the piston and connecting rod assembly. Refer to Procedure 001-054 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-054.html>)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to normal operating temperature. Check for leaks and proper operation.

Note : : All engines must be run-in after a rebuild or a repair involving the replacement of one or more piston ring sets, cylinder liners, or pistons. Reference Section 14 (Engine Testing) for the appropriate run-in procedure.

- Use the following procedure for a general run-in test overview. Refer to Procedure 014-999 in Section F. (</qs3/pubsys2/xml/en/procedures/377/377-014-999.html>)

Note : When installing new piston rings with used liners, higher than average blowby may be experienced for the first several hours of operation. If excessive blowby is still seen after several hours of engine operation, refer to the Crankcase Gases (Blowby) Excessive troubleshooting symptom tree in Section TS.

If a malfunction resulted in coolant, oil, excessive fuel, or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected.

- Use the following procedure to check the aftertreatment SCR. Refer to Procedure 014-025 in Section 14. (</qs3/pubsys2/xml/en/procedures/377/377-014-025.html>)
- Use the following procedure to check the aftertreatment DPF. Refer to Procedure 014-016 in Section 14. (</qs3/pubsys2/xml/en/procedures/493/493-014-016.html>)